

SUBJECT	Lower Secondary Maths Year 8
TOPIC	Module Seven
ASSIGNMENT	Seven
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: _____

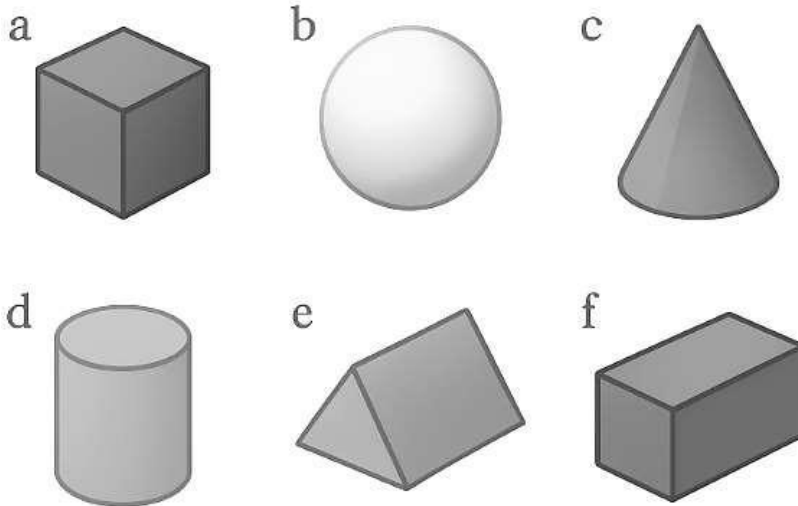
ASSIGNMENT GUIDANCE

- Make sure you write your full name clearly in the box at the top of this page.
- Please **write** your answers **by hand** in **black ink** on this question paper, unless you have obtained specific permission to submit typed work.
- This is not a timed assignment. We estimate that this assignment may take you between 30 and 40 minutes. However, if you need to take longer, then please do so.
- You are not expected to complete your assignments without referring to your notes or coursebook.
- In order for maximum marks to be awarded, for any question worth more than one mark, please show your workings to make your method clear.
- You may use a calculator in this assignment and will need a pencil and ruler.



LS Maths Assignment Seven

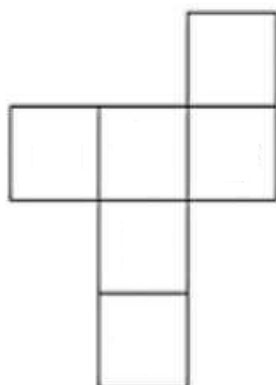
1. Match these names to the 3D objects shown below, by writing the letter of the diagram next to the name in the table.



3D object	diagram	3D object	diagram
cube		cone	
cuboid		triangular prism	
sphere		cylinder	

(6 marks)

2. Shown below is the net of a die. Number the faces from 1 to 6 so that the sum of the opposite faces is 7.



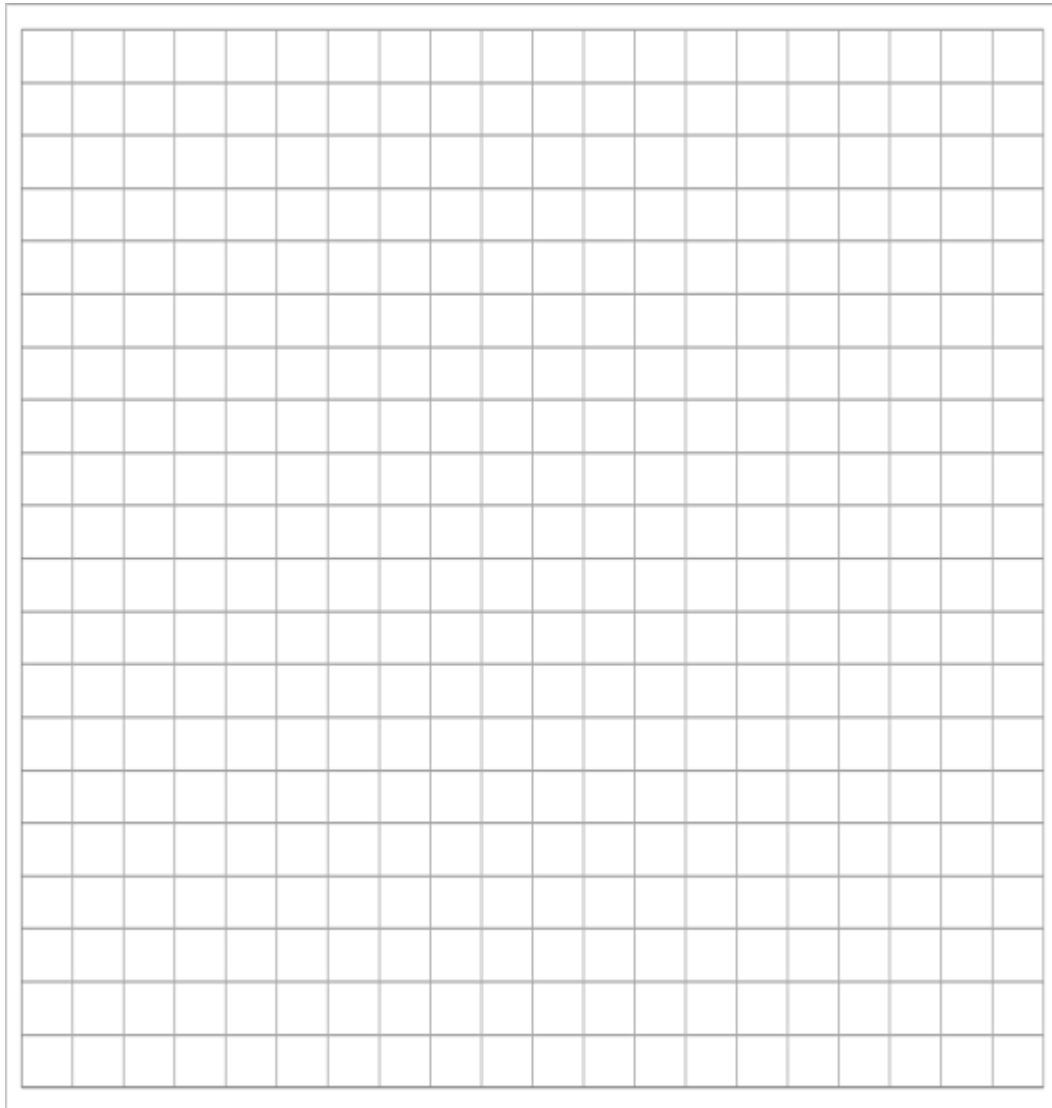
(3 marks)



3. A cuboid is 3 cm wide, 4 cm high and 5 cm long.

- a. Draw the net of the cuboid on the square paper below. Use a scale of each square representing $1\text{ cm} \times 1\text{ cm}$.

(3 marks)



- b. How many faces, vertices and edges does it have?

(3 marks)

- c. Calculate the surface area of this cuboid. You must show your workings and give units with your final answer.

(3 marks)



- d. Calculate the volume of the cuboid. You must show your workings and give units with your final answer.

(2 marks)

(Total 11 marks)

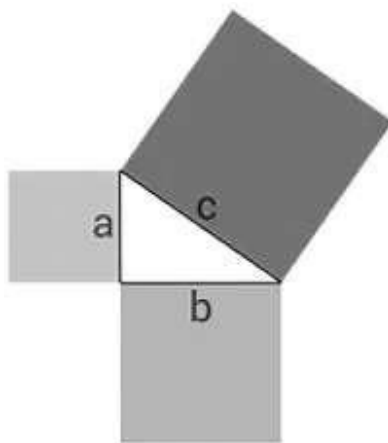
4. A cuboid is 5 cm long, 4cm wide and 3 cm high.

Tafara says that a cuboid with triple the length, width and height of this one will have triple the volume of this one.

Was Tafara correct? Calculate and compare the volumes of both cuboids and state what number the smaller cuboid needs to be multiplied by to get the volume of the larger cuboid.

(2 marks)

5. Three squares with sides of length $a = 3\text{cm}$, $b = 4\text{cm}$ and $c = 6\text{cm}$ are placed together to form a triangle as shown. **Note:** the diagram has not been drawn to scale.

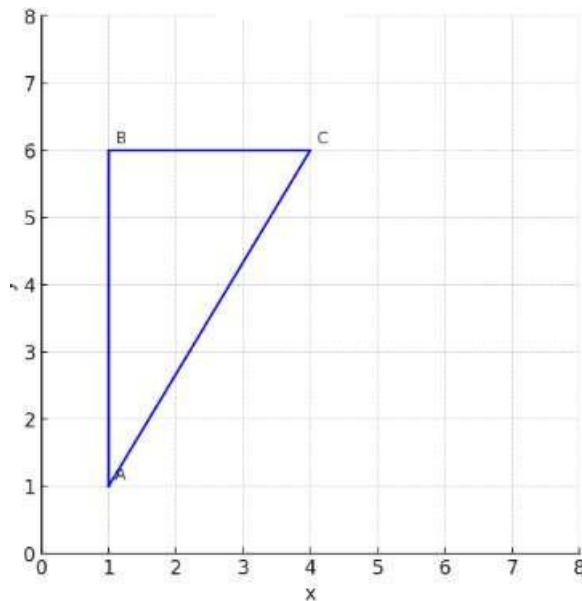


Is the triangle right-angled? Explain your answer.

(2 marks)

6. The points A(1,1) and C(4,6) are shown on the coordinate diagram.

- a. Calculate the distance AC correct to 2 decimal places.



(3 marks)

- b. Calculate the distance between the points D(-2,2) and E(5,-2) correct to 2 decimal places.

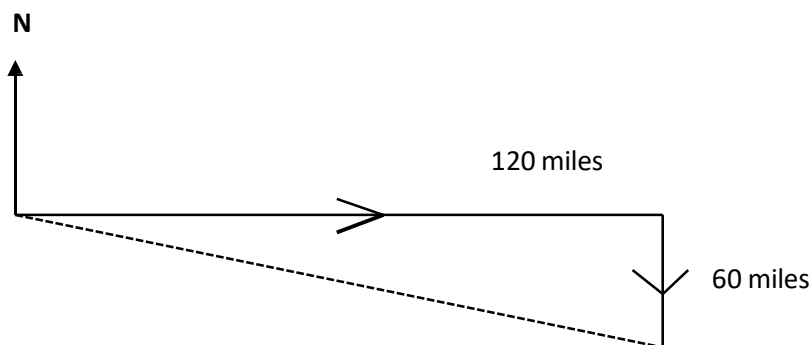
Hint: You may find a sketch helps you to use a similar method to above, but this is not essential.

(3 marks)

(Total 6 marks)

- a. From an airport, a small plane flies 120 miles east and then 60 miles south.

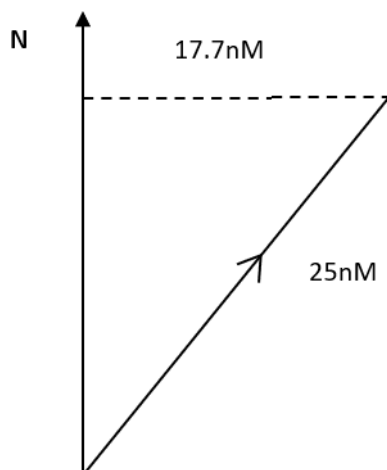
Calculate the shortest distance, back to the airfield.



(3 marks)

- b. From an island, a boat sails 25 nautical miles (nM) northeast. The boat is now 17.7 nM east of the port.

Calculate how far north of the port, in nM, the boat now is.



(3 marks)

(Total 6 marks)



8. A triangle PQR is right-angled at Q. $PQ = 9.5$ m and $QR = 6$ m.

a. Sketch this triangle.

b. Calculate the perimeter of the triangle.

(4 marks)

TOTAL FOR ASSIGNMENT 40 MARKS

